



Chapter 1

Introduction to Computers and the Internet

Internet & World Wide Web
How to Program, 5/e

<ftp://111webdesign:webdesign@120.114.190.22:4000>



Ps. 以上連結可使用網頁直接開啟 (若中文出現亂碼，按滑鼠右鍵，選擇編碼: UTF-8)
(或用ftp軟體，下載結連：<https://filezilla-project.org/>)

- ftp檔案傳輸教學 (取自網路)：<https://wiki.eztrust.com.tw/webdesign/B/10>



1.1 Introduction

- ▶ The Internet and web programming technologies are designed to be *portable* (可攜式)
 - allowing to design **web pages** and **applications** that run across an wide range of **Internet-enabled** devices.
 - 註：讓不同網際網路裝置可以執行 (如：手機、平板、筆電, etc.)

網頁程式分為：

- ▶ *Client-side* (客戶端) *programming* technologies are used to build web pages and applications that are run **on the client** (i.e., in the **browser** on the user's device).
 - 在瀏覽器端執行計算，包含：**HTML5**, **CSS3** (樣式), *JavaScript* (網頁語言), *jQuery* (*JavaScript*函式庫)等。
- ▶ *Server-side* (伺服器端) *programming*—the applications that **respond** to requests from client-side web browsers, such as **searching the Internet**, checking your **bank-account** balance, **ordering a book from Amazon**, bidding on an **eBay auction** and ordering concert tickets.
 - 在伺服器端執行計算，如：**Google search**，查詢銀行戶頭，在線上購物等



1.3 HTML5, CSS3, JavaScript, Canvas and jQuery

HTML5/6

(標籤/標記)

- ▶ **HTML** (HyperText **Markup** Language) is a special type of computer language called a *markup language* (標記式語言) designed to specify the *content* and *structure* of web pages (also called documents).
 - **HTML5**, now under development, is the emerging version (新版本) of HTML.
 - 特色：HTML5/6 希望減少瀏覽器對外掛程式的依賴：Flash、Silverlight 等(播放影片)，提供更有效、更安全的網路應用。
- ▶ HTML enables you to create content that will *display* (呈現) appropriately across the wide range of devices connected to the Internet—
 - including smartphones, tablet (平板), notebook, desktop computers.



1.3 HTML5, CSS3, JavaScript, Canvas and (cont.)

- ▶ A “**stricter**” (嚴謹的) version of HTML called **XHTML** (*Extensible HyperText Markup Language*, 可延伸超文件標示語言),
 - which is based on **XML** (eXtensible Markup Language, 可延伸標記式語言), is still used frequently today.

XML範例：

(描述資料、結構)

```
<?xml version="1.0"?>
<小紙條>
  <收件人>大元</收件人>
  <發件人>小張</發件人>
  <主題>問候</主題>
  <具體內容>早啊，飯吃了沒？ </具體內容>
</小紙條>
```

1.3 HTML5, CSS3, JavaScript, Canvas and (cont.)

Style Sheets (CSS, 層疊樣式表)

- ▶ Although HTML5 provides some capabilities for controlling a document's **presentation**, *it's better not to mix presentation with content*.
 - CSS was designed to style (樣式化) **web pages** *independently* of their content and structure.
 - 註：樣式與內容獨立(不混在一起)，才能快速更換樣式(或樣版)
- ▶ **Cascading Style Sheets (CSS)** are used to specify the *presentation*, or **styling**, of **elements** on a web page
 - (e.g., fonts/字體, spacing/間距, sizes/大小, colors/顏色, positioning/位置).
- ▶ By **separating** page styling from **page content** and structure, you can **easily change** the look and feel of the pages or a portion of a website, simply by **swapping out** (更換) one style sheet for another.
 - **CSS3** is the current version of CSS under development.



補充(網路資料)

▶ CSS1 (1994年)

- 支援字型的大小、字形
- 支援字的顏色、背景顏色
- 支援文章特徵如字母、詞和行之間的距離
- 支援文字的排列、圖像、表格
- 支援邊緣、圍框和相關排版元素

▶ CSS2 (1998年)

- 絕對的、相對的和固定的定位元素、媒體類型、雙向檔案和新的字型。
- 修改了 CSS1 和一些錯誤，刪除了不常用的內容，增加一些瀏覽器的擴充功能。

▶ CSS3 (1999年)

- CSS3 分成了不同的類別「modules」，並增加了許多功能 (如：「border-radius」、「text-shadow」、「transform」以及「transition」)。

▶ CSS4 (2011年)

- CSS4至目前功能，會視瀏覽器支援與否。

▶ CSS5 (2019年)

- 引入更多新特性，穩定增加新功能。

▶ CSS6 (2024+)

為未來CSS 所規劃功能。

1.3 HTML, CSS, JavaScript, Canvas and (cont.)

(ex: 類別 / 繼承)

JavaScript (程式語言) – 和JAVA有相似處，受Java啟發而開始設計的

- ▶ JavaScript can build **dynamic** web pages (i.e., pages that can be modified “**on the fly** (飛快/即時)” in response to *events*, such as **user input**, **time** changes).
- ▶ It enables the **client-side** (客戶端) **programming** of web applications.
- ▶ JavaScript was created by **Netscape** (網景瀏覽器, 1995~2008).
- ▶ Both Netscape and Microsoft have implemented the standardization of **ECMAScript** (**European Computer Manufacturers Association**, 歐洲計算機製造商協會)
 - 註：Netscape和Microsoft 都根據標準**ECMAScript** 實作語言，前者為**javascript**，後者為**Jscript** (兩者差異不大)。
 - The latest version of the standard is **ECMAScript 5**, corresponding to the version of **JavaScript** in this book.
- ▶ JavaScript is a **portable** (可攜式) scripting language.
 - Programs written in **JavaScript** can run in web browsers across a wide range of devices.
 - 註：“可攜式”，通常指不限於某些機器上運作。



1.3 HTML5, CSS3, JavaScript, Canvas and (cont.)

Web Browsers and Web-Browser Portability

- ▶ Ensuring a **consistent** look and feel on **client-side browsers** is one of the great **challenges** of developing web-based applications.
 - 註：在不同瀏覽器顯示相同效果，是非常**不容易**的。
(版本不同、支援性亦不同)
- ▶ Although browsers **share a common set** of features, each browser might **render** (呈現) pages differently.
 - 註：不同瀏覽器的計算/呈現技術皆不同，因此效果有些微差異。



1.3 HTML5, CSS3, JavaScript, Canvas and (cont.)

- ▶ Browsers are available in **many versions** and on **different platforms** (Microsoft Windows, Apple Macintosh, Linux, UNIX, etc.).
 - New version of browsers that may result in cross-platform **incompatibility** issues.
 - 註：不同瀏覽器版本、不同運行環境，都可能導致不相容。
- ▶ It's **difficult** to develop web pages that render correctly on all versions of each browser.
 - 註：相同網頁也無法保證在不同版本的瀏覽器顯示相同效果。
- ▶ All of the code examples in the book were tested in the **five** most popular **desktop browsers** and the **two** most popular **mobile browsers** (Fig. 1.5).
 - 註：**5**種常見的桌上型瀏覽器及**2**種行動版瀏覽器。

Most Popular **Browsers** on Desktop And Mobiles

- ▶ 看看你的瀏覽器對HTML5的支援性
 - 連結：<https://html5test.opensuse.org/>

(201x年8月資料)

Browser	Approximate market share (http://gs.statcounter.com)	Score out of 450 from html5test.com
Desktop browsers		
<i>Market share</i>		
Google Chrome 13	17%	330
Mozilla Firefox 6	27%	298
Apple Safari 5.1	7%	293
Opera 11.5	2%	286
Internet Explorer 9	40%	141
Microsoft Edge		
Mobile browsers		
<i>Mobile market share</i>		
iPhone	15% (of mobile browsers)	217
Android	18% (of mobile browsers)	184

(目前滿分571分)

Fig. 1.5 | HTML5 Test scores for the browsers used to test the examples.

1.3 HTML5, CSS3, JavaScript, Canvas and (cont.)

jQuery (Javascript函式庫)

- ▶ jQuery is currently the most popular of *JavaScript libraries*.
 - Note：目前有數以百計的Javascript相關函式庫
 - ▶ jQuery **simplifies** JavaScript programming by making it easier to **manage** a web page's elements and interact with servers in a portable manner **across** various web browsers.
 - 註：簡化javascript開發，提供更豐富互動。
 - ▶ It provides a library of **graphical user interface (GUI, 圖形介面)** controls that can be used to enhance the look and feel of your web pages.
 - 提供有別於HTML5的圖形介面，提供更好的使用者體驗。
- 舉例：jQuery 特效

(補充)

- **UI (使用者介面)**：是指產品的外觀和用戶與產品進行交互的界面設計
- **UX (使用者體驗)**：是指用戶在使用產品或服務時的整體感受，涵蓋從產品設計到使用過程的每個環節。

1.3 HTML5, CSS3, JavaScript, Canvas and (cont.)

Validating Your HTML5, CSS3 and JavaScript Code (驗證語法正確性)

▶ To ensure that browsers properly process your documents such as your **HTML5**, **CSS3** and **JavaScript** documents.

◦ 註：為了提供網頁正確運作，用於驗證語法正確性。

▶ The following validators (next page) we can use to validate the code. Where possible, we eliminated validation errors.

◦ HTML5

💡 <http://html5.validator.nu/>

💡 <http://validator.w3.org/>

◦ CSS3

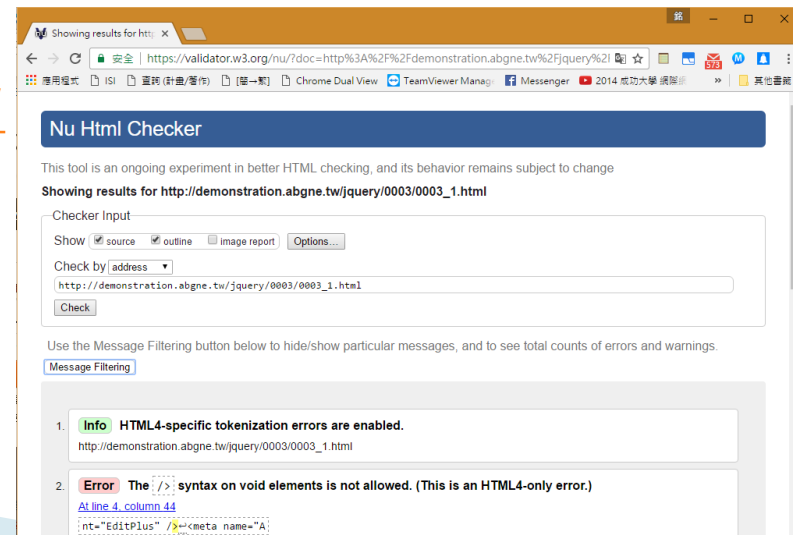
💡 <http://jigsaw.w3.org/css-validator/>

◦ JavaScript

💡 <http://www.javascriptlint.com/>

💡 <http://www.jshint.com/>

💡 測試：<https://www.nutn.edu.tw>



1.6 Web Basics

(文檔)

- ▶ A *web page* means an **HTML (HyperText Markup Language) document** (with the extension **.html** or **.htm**) that describes to a web document's content and structure.
 - 註：網頁通常指的就是.html 或.html的“文件”
 - 💡 測試看看：
chrome按右鍵 (檢視原始碼)
- ▶ **HTML documents** normally contain **hyperlinks (超連結)**, which loads a **specified web document** when **clicked**.
 - Both **images** and **text** may be **hyperlinked**.
 - 註：**hyperlinks** 表示 一檔案連接至另一檔案的“連結”。
- ▶ When **clicking** a **hyperlink**, a **web server** finds the **requested web page** and sends it to the user's **web browser**.
 - 註：將指定位置的web文件傳送至客戶端瀏覽器。
- ▶ The user can type the *address of a web page* into the **browser's address field** to view the specified page.
 - 註：在瀏覽器端，直接輸入網址



1.6 Web Basics (cont.)

- ▶ **Hyperlinks** can refer to other applications, such as e-mail addresses, files and other applications.

💡 [mailto: test@gmail.com](mailto:test@gmail.com) (連結信箱)

💡 <callTo://+8862606123> (連結skype程式)

💡 <file:///Z:/Users/Admin/Downloads/IJSE.pdf> (連結檔案)

- ▶ **Example:**

- If a **hyperlink**'s URL is in the form <mailto:emailAddress>, clicking the link loads your default e-mail program.

- Example:

💡 <mailto:test@mail.nutn.edu.tw>

- If a hyperlink refers a **file** that the browser **can not** display, it will **download** the file to store.

- Example:

💡 https://download.filezilla-project.org/client/FileZilla_3.69.6_win64_sponsored2-setup.exe

(若無法開啟連結檔，則請你下載)

1.6 Web Basics (cont.)

網路資源：

URLs and URLs ↙(最常聽到)

- ▶ **URLs** (*Uniform Resource Identifiers*, 統一資源識別元) identify resources on the Internet.
 - 註：用於描述網際網路資源的字串
- ▶ **URLs**: the URIs that start with **http://** (*Uniform Resource Locators*, 網址).

Parts of a URL

- ▶ A URL **contains information** that **directs** (指向) a browser to the **resource** to access.
- ▶ Then, **web servers** **retrieves** such resources to web clients.
- ▶ Popular **web servers** include



Apache's HTTP Server

(下載網址：<https://httpd.apache.org/>)



Microsoft's Internet Information Services (IIS)

(下載網址：<https://www.microsoft.com/zh-TW/download/details.aspx?id=48264>)

(教學/win10內安裝) <https://me1237guy.pixnet.net/blog/post/62600393>



APACHE

註：兩種架設**web**服務的應用程式：Apache、IIS等 (其他類似，如：[Tomcat](#))

Making a Request and Receiving a Response

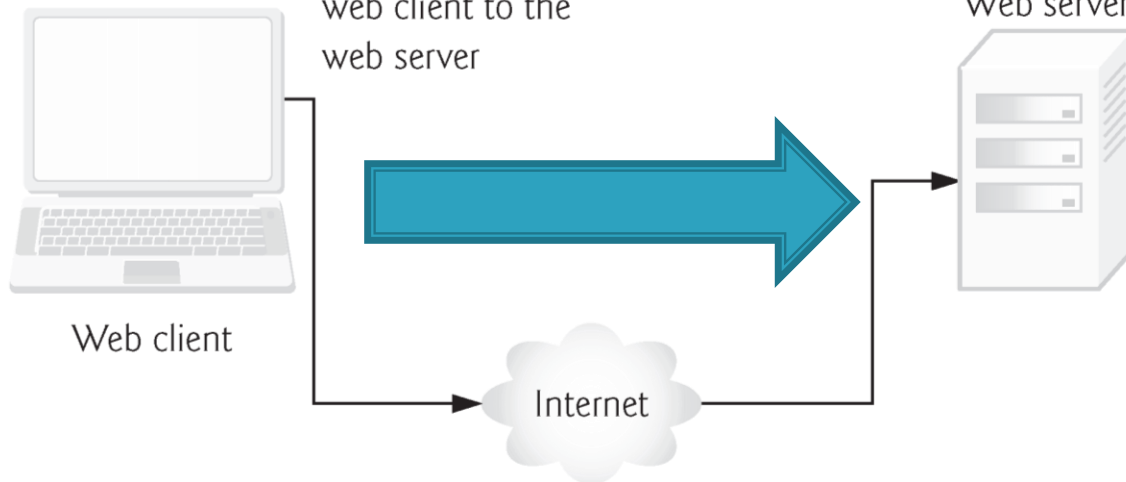


▶ Example:

- **Request (請求):** 瀏覽器向web server索取一個web documents.

(步驟1:

Client 發出請求
，檢閱某份文檔)



- b) After it receives the request, the web server searches its system for the resource

(步驟2:

web server 收到請求
，查看是否有此文檔)

Fig. 1.8 | Client requesting a resource from a web server.



1.6 Web Basics (cont.)

- ▶ The web browser **sends** an **HTTP request** to the server.
The **request form** (請求格式) is

固定格式: **GET** **/books/downloads.html** **HTTP/1.1**

命令:“取回” 文件位置 傳輸協定(及版本)

- 網路爬蟲：
透過linux平台下指令抓html
(例如：比價網)

- ▶ The word “**GET**” is an **HTTP method**
 - that the client **requests** a **resource** (document or file) from the server.
- ▶ **/books/downloads.html**
 - The **path name** of the resource (e.g., an HTML5 document)
- ▶ **HTTP/1.1**
 - The **protocol** (協定)’s **name** and version number

Making a Request and Receiving a Response



► Example:

- **Response (回覆):** web server回覆client browser指定的文檔

(步驟2:

Client接收回覆的文檔)

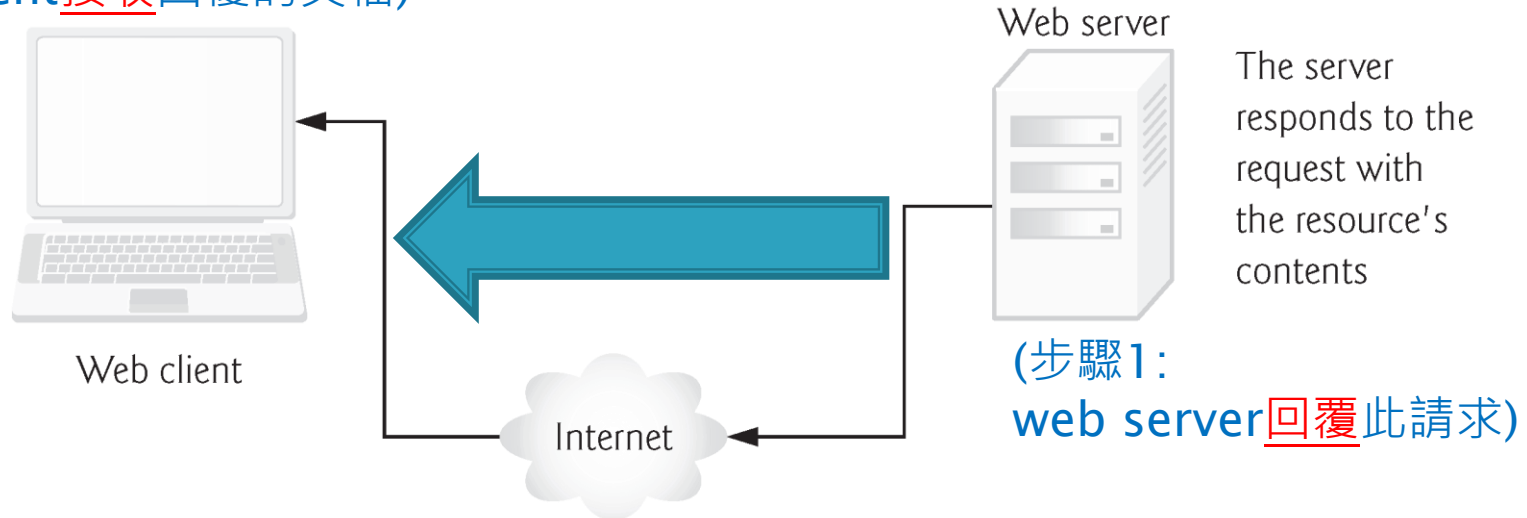


Fig. 1.9 | Client receiving a response from the web server.

1.6 Web Basics (cont.)

Web server response (server端回覆方式)

- ▶ The **server** sends a **text** that indicates the **HTTP version** and a **numeric code** and the status of the transaction.
 - For example, indicates **success** (表示成功)

固定格式: HTTP/1.1 200 OK
 傳輸協定版本 編號 狀態

HTTP/1.1 404 Not found

傳輸協定版本 編號 狀態

- ▶ informs the client that the requested resource is not found.

HTTP Headers (HTTP標頭檔: 描述內容類型)

- In addition, the server sends the HTTP headers to provide more information about the data that will be sent.
- Example: Content-type: text/html 或 Content-type: image/jpeg
 內容類型 Html文檔 內容類型 jpeg圖片

- 網路爬蟲：
透過linux平台下指令抓html
(例如：比價網)

1.6 Web Basics (cont.)

HTTP “get” and “post” Requests

- ▶ The two most common HTTP request types (also known as request methods) are get and post.
 - 註：兩種HTTP詢求方式：get 和 post.
- ▶ **get**: retrieves (取回) information from a server, such as an HTML document, an image or search results (例如：google查詢).
 - 註：索取一個HTML文檔、圖片、或查詢結果。
- ▶ **post**: sends (張貼/送出) data to a server (例如：張貼表單).
- ▶ For example: (差別: 網址是否出現)
 - **Get** requests and **post** requests can be used to send data to a web server, but request type are different.
 - “Google search”: the web server receives the key words as part of the request.
 - 💡 比如：在Google網頁上，搜尋：“html5 demo”
 - <https://www.google.com.tw/search?q=html5+demo>

1.6 Web Basics (cont.)



- **get** request:
- <https://www.google.com.tw/search?q=html5+demo&ie=UTF-8>
- 編碼

特定意義: 負責處理的接口 檢索之變數 變數之內容

- “search” is Google’s server-side form handler
- “q” is the variable in Google’s search form
- “html5+demo” is the search term.
- “?” separates (區隔) the query string from the content of the URL.
- “=” separates the variable and its value.
- “&” separates the different variable (變數) and their values (內容).

1.6 Web Basics (cont.)

- ▶ **post request**: 夾帶在HTTP訊息中(非網址上)
 - Can send **large amounts of information**.
 - Can hide the **submitted data** in an HTTP message.

Example:

A screenshot of a web browser window. The address bar shows the URL `http://www.php5.idv.tw/php5-article/ch1/ch1-06/form3.php`. The page title is "PHP表單傳遞實例3". The main content area displays the text "表單傳來了變數:" followed by a horizontal line. Below this is the prompt "請輸入想買的東西:" and a text input field containing the word "水果". At the bottom, there are two buttons: "買東西" and "不買".



A screenshot of the same web browser window after the form has been submitted. The address bar and page title remain the same. The main content area now displays "表單傳來了變數:" followed by the text `item=水果submit=買東西`. Below this, it says "你買了:水果" and "謝謝光臨". The input field now contains "奶粉".

1.7 Multitier Application Architecture

- ▶ Web-based applications are often multitier applications
 - Sometimes referring to *n*-tier applications that divide functionality into separate tiers註：web應用會根據“功能”切分成多個層次
- ▶ The following shows a three-tier web-based application.
 - Top tier: gathers **input** (輸入) and **displays** (呈現) output.
 - Middle tier: ensures that data is **reliable** (可靠).
 - Bottom tier: maintains the application's **data** (維護資料).

(頂層: 又稱“client層”)

(中間層: 確保資料可靠)

(底層: 又稱“資料層”)

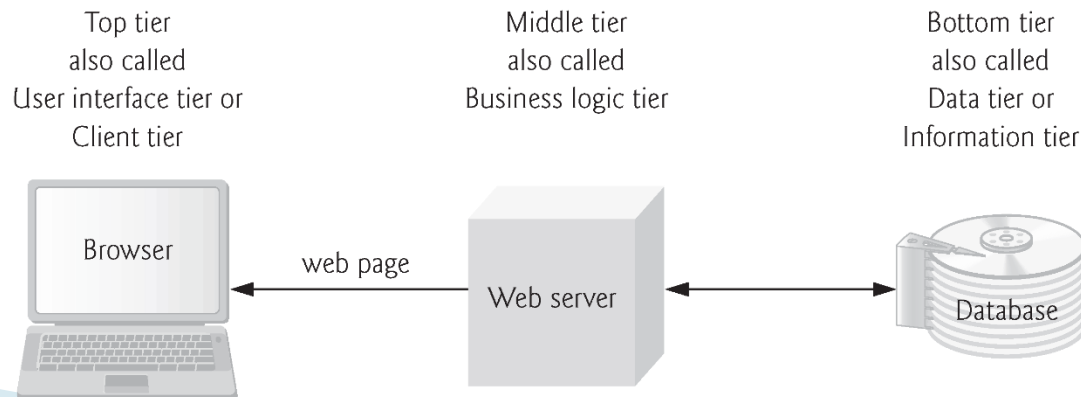


Fig. 1.10 | Three-tier architecture.

1.8 Client-Side Scripting v.s. Server-Side Scripting



- ▶ (客戶端/前端) ^(腳本/文檔) **Client-side scripting** with **JavaScript** is used to
 - to validate **user input** (ex: 帳號字母大小寫/數字/長度, etc.)
 - to **interact** with the browser (ex: 翻頁、拖拉、滑動, etc.)
 - to enhance web pages' look and feel (ex: 動畫、花俏)
 - to add client/server communication between a browser and a web server.
- ▶ Limitations of **Client-side scripting**:
 - **browser dependency** (瀏覽器本身也要支援script才行)
 - **Can not** arbitrarily **access** the **local hardware** and **file system** for security reasons. (不允許任意存取本機硬體及檔案)
 - For sensitive information (e.g., **passwords** or personally identifiable data), **should not** be on the client.
- ▶ (伺服器/後端) **Server-side scripts**
 - more flexibility (針對client端客製化回應)
 - 例子1) : Google搜尋指定“關鍵字”的網頁
 - 例子2) : 進入航空公司網頁，查詢某航班時間資訊，經由server查詢航班資料庫，動態產生包含該資料的HTML文檔，再送回client端。



1.9 World Wide Web Consortium (W3C)

- ▶ October 1994
 - (開放性組織) Tim Berners-Lee founded an organization—the World Wide Web Consortium (W3C)—
 - to develop nonproprietary (非專有), interoperable technologies (協同工作技術) for the World Wide Web.
- ▶ Primary goals of W3c
 - to make the web *universally accessible* (全世界皆可存取) — regardless of language or culture.
- ▶ The W3C is also a standards organization (定訂web的標準協議).
- ▶ Web technologies standardized by the W3C are called Recommendations.
 - which is a document that specifies a technology's role and syntax (語法).
 - Current Recommendations include
 - Hyper Text Markup Language 5 (HTML5),
 - Cascading Style Sheets 3 (CSS3),
 - Extensible Markup Language (XML).



HTML5 簡易展示

1. <https://www.bootstrapmb.com/tag/html5texiao>
2. <https://codepen.io/jackrugile/pen/kMWyeM>
3. <https://codepen.io/towc/pen/mJzOWJ>
4. https://codepen.io/mladen___/pen/gbvqBo
5. <https://webdesign.tutsplus.com/21-ridiculously-impressive-html5-canvas-experiments--net-14210a>

課本提供的Demo網址

URL	Description
http://alteredqualia.com/canvasmol/	Uses canvas to display a 3D molecule that can be viewed from any desired angle (0–360 degrees).
http://pasjans-online.pl/	The game of Solitaire built using HTML5.
http://andrew-hoyer.com/experiments/cloth/	Uses canvas to simulate of the movement of a piece of cloth. Click and drag the mouse to move the fabric.
http://www.paulrhayes.com/experiments/cube-3d/	CSS3 demo allows you to use the mouse to tilt and rotate the 3D cube. Includes a tutorial.
http://www.effectgames.com/demos/canvascycle/	Animated waterfall provides a nice demo of using color in HTML5 canvas.
http://macek.github.com/google_pacman/	The Google PAC-MAN® game (a Google Doodle) built in HTML5.
http://www.benjoffe.com/code/games/torus/	A 3D game similar to Tetris® built with JavaScript and canvas.
http://code.almeros.com/code-examples/water-effect-canvas/	Uses canvas and JavaScript to create a water rippling effect. Hover the cursor over the canvas to see the effect. The site includes a tutorial.
http://jqueryui.com/demos/	Numerous jQuery demos, including animations, transitions, color, interactions and more.
http://lab.smashup.it/flip/	Demonstrates a flip box using jQuery.
http://tutorialzine.com/2010/09/html5-canvas-slideshow-jquery/	Slideshow built with HTML5 canvas and jQuery (includes a tutorial).
http://css-tricks.com/examples/Circulate/	Learn how to create an animated circulation effect using jQuery.
http://demo.tutorialzine.com/2010/02/photo-shoot-css-jquery/demo.html	Uses jQuery and CSS to create a photoshoot effect, allowing you to focus on an area of the page and snap a picture (includes a tutorial).

Fig. 1.7 | HTML5, CSS3, JavaScript, canvas and jQuery demos. (Part 2 of 2.)



LAB 1

Lab 1: 比較瀏覽器對HTML5的支援性

目標 (大家來找碴) :

1) 試著安裝下列不同桌上型瀏覽器(挑2種) , 並比較瀏覽器對HTML5之支援性 , 列出3項呈現有相異的HTML5網站

◦ 【注意】繳交方式: 將差異處截圖後儲存於word檔 , 註明 :

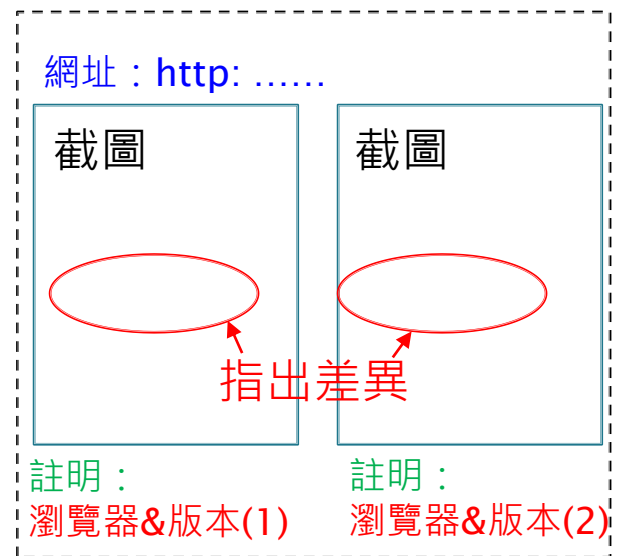
- 1) 網址
- 2) 瀏覽器 & 版本
- 3) 差異處(說明)

▶ 參考資料 :

◦ 桌上型瀏覽器 :

- Chrome
下載 : <https://www.google.com.tw/chrome/>
- Firefox
下載 : <https://www.mozilla.org/zh-TW/>
- Opera
下載 : <https://www.opera.com/zh-tw>
- Internet Explorer (內建)
- Safari (Apple macOS)

Word 繳交範例:



▶ 2) Bonus (加分題)

- 1. 根據網頁validator (講義第13頁) , 試著檢查2個網站 , 並截圖記下error或warning儲存於word檔。
- 2. 試著舉出使用get和post用法的網址 (各2個) , 並截圖儲存於word檔。

HTML5 相關網站



▶ [HTML5 有趣特效](#)

▶ [HTML5 樣板網站](#)



▶ [Chromeexperiments.com](#) (有趣例子)

▶ [Michaeljared.itch.io/a-beavers-tale](#) (小遊戲)

▶ 等等.. 也可以自行搜尋”**html5 特效**”，結果會有很多。

- 注意：

- 請於**23:59**前上傳至：

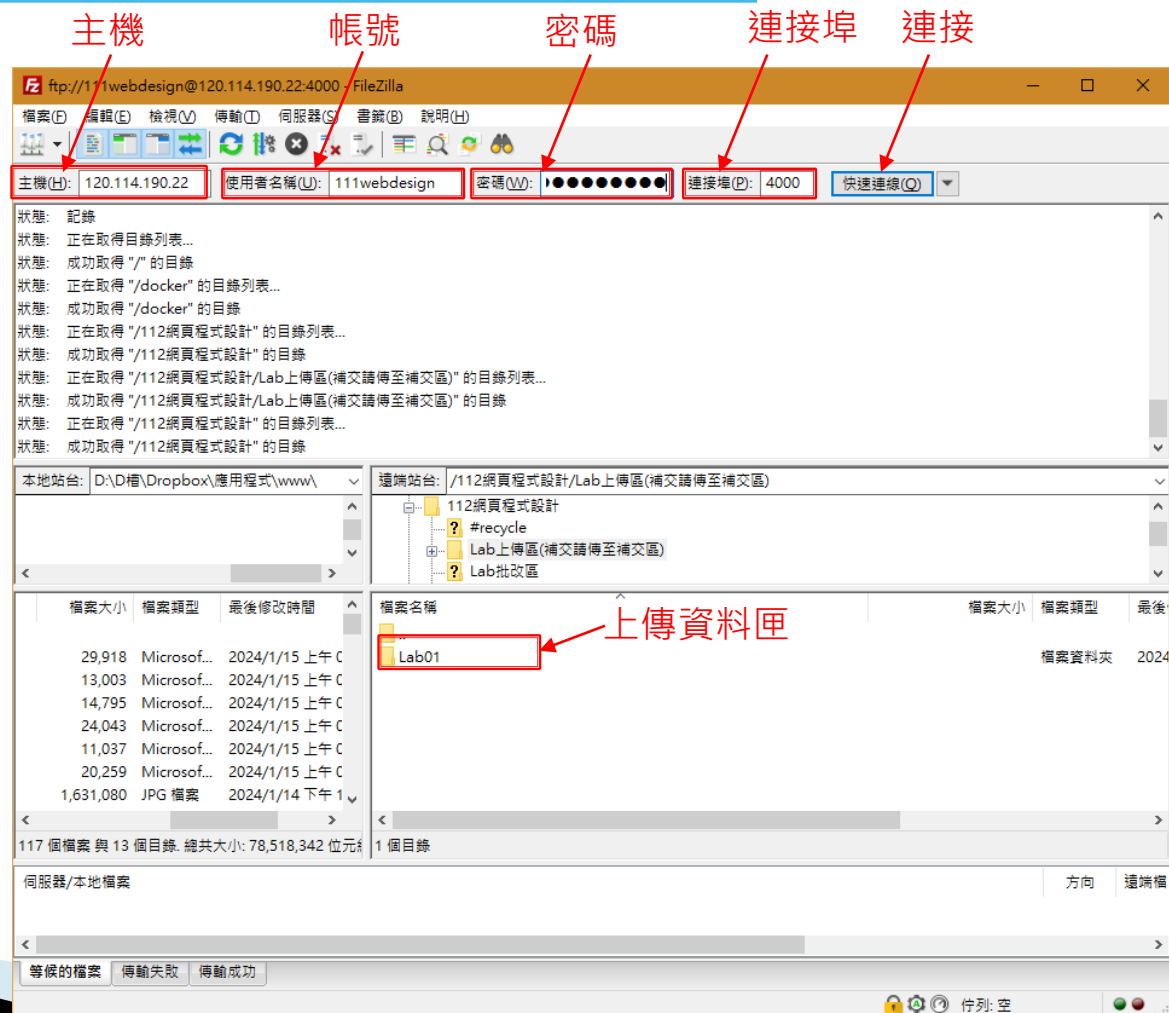
- <ftp://111webdesign:webdesign@120.114.190.22:4000>

- 請以”學號”建立資料匣 (**否則扣分**)

ftp檔案傳輸教學 (取自網路)：

▶ 教學：<https://wiki.eztrust.com.tw/webdesign/B/10>

- ▶ 主機：120.114.190.22
- ▶ 連接埠：4000
- ▶ 帳號：111webdesign
- ▶ 密碼：webdesign





課輔群組 / 補救教學

(每週三13:00-15:00)

- 掃描QRcode，加入課輔群組~!

113-2 網頁程式設計 上課座位表 (簽名)



課輔群組(加入)

____年 ____月 ____日

後 門				